

3a	When two (or more) waves overlap, the resultant displacement (at any point and instant) is the (vector) sum of the displacements due to each individual wave (at that point and instant).	2
3bi	$\frac{5}{34} \times 2\pi = 0.924 \text{ rad}$	2
3bii	Amplitude of maxima = $2.4 + 1.6 = 4.0$ units Amplitude of minima = $2.4 - 1.6 = 0.8$ units $= \frac{4.0^2}{0.8^2}$ Ratio of intensity = 25	1 1 1
3c	Path difference at Q = $\frac{6.5}{4.0} \lambda = 1.625\lambda$ Path difference at A = $\frac{27}{4.0} \lambda = 6.75\lambda$ Along AQ, maxima are where path difference is $2\lambda, 3\lambda, 4\lambda, 5\lambda$ and 6λ. So 5 maxima.	1 1
3d	Even though the two waves are travelling in opposite directions, the amplitude is different. Thus no stationary wave is formed.	1 1

2021

HWA CHONG INSTITUTION (COLLEGE SECTION)

C2

4a

By Newton's Second Law, taking upwards as positive